Lesson plan · for the teacher

Safe Wi-Fi at school, at home and in town

Cyber security · Level 2 · Lesson 4

Time
40 min

Materials
Board and chalk; An open sheet of paper (a postcard) and a sealed envelope; A phone to show a website's address bar (optional)

Learning goals

- Explain what the owner of a Wi-Fi network can see.
- Explain what the padlock (HTTPS) hides, and what it doesn't prove.
- Choose when to use mobile data instead of free Wi-Fi.

Activity: What can the café owner see?: Postcards and envelopes

1. Warm-up (5 min): Hold up an open sheet of paper with a short message on it. Ask: if this travelled by post, who could read it on the way?
2. Teach (8 min): Compare a plain HTTP page to a postcard, and HTTPS to a sealed envelope with only the address on the outside. Explain that the café's Wi-Fi is like a post office everything passes through.
3. Draw the journey (15 min): Pupils copy the café, the phone and the website from the board, draw the messages between them, and mark what the café owner can read with and without the padlock.
4. Share (7 min): Groups agree on one thing never to do on free Wi-Fi and explain why. Write the class rule: money and marks over mobile data.
5. Wrap-up (5 min): Exit ticket: each pupil writes one thing they will never do on free Wi-Fi, and why.

Before the lesson, draw a café, a phone and a website on the board. Prepare an open sheet with a short message (the postcard) and a folded, sealed envelope with only an address on it.

Discussion

- What can the café owner see on a plain page?
- Does the padlock prove a website is honest?
- Why should money go over mobile data?

Exit ticket

- One thing I will never do on free Wi-Fi + why.

Try it at school tomorrow
Draw a café, a phone and a website on the board. Pupils draw the messages travelling between them and mark what the café owner can read: everything on a plain page, only the site's name with the padlock (HTTPS). Finish with the class rule: money and marks over mobile data.

Practice questions · for you

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1 **The new school router still uses the admin password it came with. Is it safe to leave it?**

- ◯ Yes: only staff ever see the router
- ◯ Yes, as long as the Wi-Fi itself has a password
- ◯ No: change it, because the passwords routers come with are often easy to find or guess

2 **You're on a café's free Wi-Fi. Which of these are fine to do there?**

- ☐ Pay a bill with mobile money
- ☐ Watch a lesson video
- ☐ Sign in to the school's marks system
- ☐ Read the news
- ☐ Look up tomorrow's weather

3 **If a page shows the padlock (HTTPS), the website must be honest.**

- ◯ True
- ◯ False

4 **You realise you typed your email password on a fake hotspot's sign-in page. Put what to do in order.**

- ☐ Switch to mobile data
- ☐ Check recent sign-ins and sign out other devices
- ☐ Disconnect from the network
- ☐ Change your email password

5 **Match each situation to the sensible choice.**

A café, reading the news	• • • •	Mobile data
A bus station, paying a bill	• • • •	The free Wi-Fi is fine
Home, a new router	• • • •	A guest network
School, visitors at an event	• • • •	A new admin password and WPA2 or WPA3

✎ **Where do you most often use Wi-Fi that isn't yours? What will you do differently there from now on?**

Check your answers

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1 **No: change it, because the passwords routers come with are often easy to find or guess**

The admin password controls the whole router, and the ones routers come with are often listed online or easy to guess.

2 **Read the news; Watch a lesson video; Look up tomorrow's weather**

Reading, watching and looking things up share nothing sensitive.

3 **False**

The padlock means the connection is private, so the café owner can't read it.

4 **Top to bottom: 2, 4, 1, 3**

Leave the fake network first, then use a connection you trust to change the password, then look for anyone who already got in.

5 **A café, reading the news — The free Wi-Fi is fine**
A bus station, paying a bill — Mobile data
Home, a new router — A new admin password and WPA2 or WPA3
School, visitors at an event — A guest network

Match the protection to what's at stake: reading is fine anywhere, money needs your own connection, and your own networks need locks and separation.